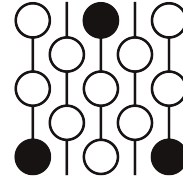


Review of Submission 2A

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The AI got the problem correct. This is a working proof, with all essential details included. The proof in both halves is efficient, and the proof in the second half is especially clean. I really enjoyed reading the second half.

Novelty. Let me first discuss the context for the problem and then describe how the AI did. The problem is based on my solution of the Halpern-Weaver Conjecture, a result which asks for the infimal aspect ratio needed for a smooth and isometric embedding of a flat Moebius band into $\mathbb{R}^3$. My paper was published in the *Annals* in January, 2025 and there are also some of accounts of the result in the popular media. The problem the AI considered is the same problem in a different category of maps, piecewise linear maps with a certain side condition called squeezing.

The first half of the problem amounts to showing that $\sqrt{3}$ is a lower bound on the allowable aspect ratio. For this, the AI seems to have followed my paper very closely, adapting essentially every one of my arguments to the new category of mappings. One thing that troubled me is that the AI did not cite my work at all, even though the arguments were extremely close. Indeed, the AI used the same terminology in my paper – e.g. "T-patterns" and "bends" – and also used the same letters, B, T, D, H for the important objects in the proof. The chances of this happening without the AI actually reading and absorbing my paper seem to be a trillion to one against. If I found this kind of thing written on a student's exam – close adaptation without any acknowledgement – I would probably give them an F for the exam.

The second half of the problem amounts to constructing an example for every aspect ratio greater than $\sqrt{3}$. For this, the AI did a beautiful and, as far as I can tell, original job. The AI produced a 6-triangle example which works for a wide range of parameter values in the one-sided neighborhood of $\sqrt{3}$. (For large aspect ratio the result is obvious.) This construction was both cool and suprising, just great.

2 Recommendation

I would recommend that the paper is publishable with minor revisions, though with a big *caveat*. I will indicate the revisions with inline comments below. Now let me explain the one *caveat*. As

indicated above, the close adaptation of my arguments without any citation was annoying. I would consider this a kind of ethical failing in a human author. For an AI, it is hard to tell. Possibly I would reject the paper on the grounds of “unethical AI behavior”, whatever that might mean.

3 The Solution

The infimum is $\sqrt{3}$

Let $M_\beta = \Sigma / \langle G_\beta \rangle$. This is a compact flat Mobius band, with boundary a circle of length 2β . If β is realized, condition (2) says exactly that the given map is constant on the G_β -orbits and separates distinct orbits. Hence it descends to a continuous injection $F : M_\beta \rightarrow \mathbb{R}^3$, and this injection is an embedding because M_β is compact and $\mathbb{R}^3$ is Hausdorff. The locally finite invariant triangulation descends to a locally finite triangulation of the compact surface M_β , hence to a finite one.

Reviewer 1

The term bend comes from my paper on the optimal paper Moebius band, which this paper should cite.

In a clean triangle, call the boundary edge the ridge and the opposite vertex the apex. A bend is a segment in the triangle joining the apex to a point of the ridge.

Lemma 1 (bends lengthen). *Let L be the affine map on one clean triangle. If P is any point of the ridge and C is the apex, then*

$$|L(P) - L(C)| > |P - C|.$$

Moreover every non-degenerate boundary subarc is strictly shortened by F .

Reviewer 1

This proof can be done without calculus, but this here is still a decent proof.

Proof. Let the ridge endpoints be A, B , and put $P_s = (1 - s)A + sB$. The function

$$q(s) = |L(P_s) - L(C)|^2 - |P_s - C|^2$$

satisfies

$$q''(s) = 2(|L(A) - L(B)|^2 - |A - B|^2) < 0,$$

because the ridge is shortened. Since the two side edges are lengthened, $q(0) > 0$ and $q(1) > 0$. Concavity gives $q(s) > 0$ for $0 \leq s \leq 1$. On each boundary edge the affine map has constant Lipschitz factor < 1 ; an arbitrary boundary subarc is split at the finitely many boundary vertices it meets, and the strict inequalities add. $\square$

Reviewer 1

This is Lemma T in my paper on the optimal paper Moebius band. That work should be cited, because the proof is extremely similar.

Lemma 2 (a polygonal T -pair). *For $F(M_\beta)$ there are two bends whose image segments lie on perpendicular intersecting lines.*

Proof. The bends form a circle. Indeed, every bend meets the center circle $\{y = 1/2\}/\langle G_\beta \rangle$ once, and every point of the center circle lies in a unique bend; on a common non-boundary edge the two adjacent triangles give the same limiting bend.

Parametrize this bend circle by $u \in \mathbb{R}/\mathbb{Z}$, lift to $u \in \mathbb{R}$, and choose the image direction $e(u)$ from the lower side of the lifted strip to the upper side. Let $m(u)$ be the image midpoint. The previous lemma makes every image bend non-degenerate, and the finite polygonal structure gives continuous functions e, m . Going once around the Mobius band reverses the chosen orientation, so

$$e(u+1) = -e(u), \quad m(u+1) = m(u).$$

Reviewer 1

This proof seems too terse for me. In particular, the functions are not motivated at all. They work, but the reader might want some geometric intuition for what is going on.

For $0 < r < 1$ define

$$\Phi(u, r) = \left(e(u) \cdot e(u+r), (m(u) - m(u+r)) \cdot (e(u) \times e(u+r)) \right).$$

It is periodic in u . As $r \rightarrow 0$ it tends to $(1, 0)$, and as $r \rightarrow 1$ it tends to $(-1, 0)$. Thus it extends continuously to the two-point compactification of $S^1 \times (0, 1)$, obtained by collapsing the two boundary circles of $S^1 \times [0, 1]$; this compactification is S^2 .

The involution $\iota(u, r) = (u+r, 1-r)$ interchanges the two bends. Under the coordinates

$$(\sin \pi r \cos 2\pi(u+r/2), \sin \pi r \sin 2\pi(u+r/2), \cos \pi r)$$

on the compactified cylinder, ι is the antipodal map. From $e(u+1) = -e(u)$ and $m(u+1) = m(u)$ one checks that $\Phi \circ \iota = -\Phi$. Borsuk–Ulam gives a zero of Φ .

At such a zero the two direction vectors are perpendicular. The second coordinate says that the vector joining the two midpoints lies in the plane spanned by those directions; hence the two supporting lines are coplanar. Since they are not parallel, they intersect. $\square$

Reviewer 1

This is the same as the main geometric argument in my paper. The logic is very similar, and the labels T, B, D, H are as in my paper. This should be cited.

Proposition 1. *Every realized β satisfies $\beta > \sqrt{3}$.*

Proof. Choose the two bends from Lemma 2. Their image segments have disjoint relative interiors, because F is an embedding. Let O be the intersection of their supporting lines. At least one image segment does not contain O in its relative interior; call that bend B , and call the other one T .

Lift T to a segment from $(0, 0)$ to $(t, 1)$. Replacing x by $-x$, if necessary, we may assume $t \geq 0$. In fact $t < \beta$: if $t > \beta$, this lifted segment meets its G_β -translate in its interior, and if $t = \beta$, its endpoints are G_β -equivalent. Either case contradicts the fact that the image of a bend is a non-degenerate affine segment and that F is embedded on the quotient.

Cutting M_β along T gives two boundary arcs H, D of lengths

$$\text{length}(H) = \beta - t, \quad \text{length}(D) = \beta + t. \quad (1)$$

Indeed the upper endpoint $(t, 1)$ has boundary coordinate $t + \beta$ modulo 2β . Let $T^* = F(T)$. Since $H^* = F(H)$ joins the endpoints of T^* , while boundary arcs are shortened and bends are lengthened,

$$\beta - t = \text{length}(H) > \text{length}(H^*) \geq |T^*| > \sqrt{1 + t^2}. \quad (2)$$

The line of B^* is perpendicular to the line of T^* . Since O is not in the relative interior of B^* , one endpoint v^* of B^* has distance from the line of T^* at least $|B^*|$. The domain length of B is at least 1, so Lemma 1 gives $|B^*| > 1$. Let A be the boundary arc, either H or D , which contains the corresponding boundary endpoint v .

The image arc A^* goes from one endpoint of T^* to the other through v^* . If a point has perpendicular distance h from a line segment of length L , the sum of its distances to the two endpoints is at least $\sqrt{L^2 + 4h^2}$, with equality only over the midpoint. Here $L = |T^*| > \sqrt{1 + t^2}$ and $h > 1$, so

$$\text{length}(A) > \text{length}(A^*) \geq (\text{that sum}) > \sqrt{5 + t^2}. \quad (3)$$

Reviewer 1

This endgame seems a bit inefficient to me. It works, but is not maximally efficient. The main point is to get $\beta > \max(\sqrt{1 + t^2} + t, \sqrt{5 + t^2} - t)$.

If $A = H$, then $D \geq H$ and (3) gives

$$2\beta = H + D > 2\sqrt{5 + t^2} > 2\sqrt{3}.$$

Thus assume $A = D$. From (1)–(3),

$$2\beta = H + D > \sqrt{1 + t^2} + \sqrt{5 + t^2}, \quad (4)$$

and also

$$2\beta = 2D - 2t > 2\sqrt{5 + t^2} - 2t. \quad (5)$$

Let

$$a(t) = \sqrt{1 + t^2} + \sqrt{5 + t^2}, \quad b(t) = 2\sqrt{5 + t^2} - 2t.$$

On $[0, \infty)$, a is increasing and b is decreasing, and

$$a(1/\sqrt{3}) = b(1/\sqrt{3}) = 2\sqrt{3}.$$

Hence $\max(a(t), b(t)) \geq 2\sqrt{3}$ for all $t \geq 0$. Equations (4)–(5) imply $2\beta > 2\sqrt{3}$. $\square$

It remains to construct realizations for all larger β .

Reviewer 1

In general, this second half is really cool and interesting. I was surprised that you could make one of these things with just 6 triangles. My own construction, which is more transparent, used 11.

Lemma 3 (an embedded six-triangle model). *Let $E_0E_1E_2$ be an equilateral triangle of side a . For every sufficiently small $\varepsilon > 0$ there are points $C_i(\varepsilon)$ tending to the midpoint of E_iE_{i+1} (indices mod 3) such that the six affine triangles*

$$(E_i, C_i, E_{i+2}), \quad (C_i, E_{i+1}, E_{i+2}), \quad i = 0, 1, 2,$$

form an embedded polyhedral Mobius band with boundary $E_0C_0E_1C_1E_2C_2E_0$.

Reviewer 1

A diagram for this proof would be extremely helpful. The diagram ideally would show both the domain and the range.

Reviewer 1

The notation switch here from C,E to A,B is a bit confusing. Logically it makes sense, but a “connecting comment” would be nice.

Proof. Embeddedness is preserved by nonsingular affine changes of coordinates, so it is enough to verify one model. Take

$$\begin{aligned} A_0 &= (0, 0, 0), & A_1 &= (1, 0, 0), & A_2 &= (0, 1, 0), \\ B_0 &= (1/2, 0, \varepsilon), & B_1 &= (1/2 - \varepsilon, 1/2, -2\varepsilon), & B_2 &= (\varepsilon, 1/2, 0), \end{aligned}$$

where $0 < \varepsilon < 1/4$, and take the six faces

$$\begin{aligned} \Delta_0 &= A_0B_0A_2, & \Delta_1 &= B_0A_1A_2, & \Delta_2 &= A_1B_1A_0, & \Delta_3 &= B_1A_2A_0, \\ \Delta_4 &= A_2B_2A_1, & \Delta_5 &= B_2A_0A_1. \end{aligned}$$

Reviewer 1

This is a really wierd and confusing notation for the links which, after a lot of head-scratching, I figured out. In hindsight, I sort of like the notation, but some explanation might be nice. Perhaps $\leftrightarrow$ is better than what looks like minus sign. Again, a diagram would be very helpful here.

The links of the vertices are intervals. For instance

$$\text{lk}(A_i) = B_i - A_{i+2} - B_{i+1} - A_{i+1} - B_{i+2}, \quad \text{lk}(B_i) = A_i - A_{i+2} - A_{i+1}$$

(indices mod 3, and reversal of a path is immaterial). Thus the abstract complex is a connected compact surface. Its single boundary cycle is $A_0B_0A_1B_1A_2B_2A_0$, and $\chi = 6 - 12 + 6 = 0$; hence, by the classification of compact surfaces, it is a Mobius band once its geometric realization is embedded.

Reviewer 1

I like this proof a lot, though I did not try to follow it line by line. Instead I plotted a picture, with specific choices, in Mathematica, and saw that it was embedded. I did this by checking every pair of triangles visually.

It remains to check embeddedness. The face normals are

$$\begin{aligned} n_0 &= (-\varepsilon, 0, 1/2), & n_1 &= (\varepsilon, \varepsilon, 1/2), & n_2 &= (0, 2\varepsilon, 1/2), \\ n_3 &= (2\varepsilon, 0, 1/2 - \varepsilon), & n_4 &= (0, 0, 1/2 - \varepsilon), & n_5 &= (0, 0, 1/2). \end{aligned}$$

For the edge-sharing pairs $(0, 1), (0, 3), (1, 4), (2, 3), (2, 5)$ these normals are not proportional, so the two planes meet in the line of the common edge and the two triangles meet exactly in that edge. The remaining edge-sharing pair is $(4, 5)$. These two triangles are coplanar in $z = 0$, but their third vertices lie on opposite sides of their common edge B_2A_1 , since

$$\det(A_1 - B_2, A_2 - B_2) = \frac{1 - 2\varepsilon}{2} > 0, \quad \det(A_1 - B_2, A_0 - B_2) = -\frac{1}{2} < 0.$$

Therefore $\Delta_4 \cap \Delta_5 = B_2A_1$.

The other nine pairs of faces share only one vertex. For each of them, the normals listed above are not proportional, so the two planes meet in a line through that vertex. Expressing one direction of this line in the two edge bases at the shared vertex gives the following sign table:

(i, j)	Δ_i	Δ_j
$(0, 2)$	$(-, +)$	$(-, +)$
$(0, 4)$	$(-, 0)$	$(-, +)$
$(0, 5)$	$(0, -)$	$(-, +)$
$(1, 2)$	$(-, +)$	$(+, -)$
$(1, 3)$	$(+, -)$	$(-, +)$
$(1, 5)$	$(0, +)$	$(+, -)$
$(2, 4)$	$(0, -)$	$(+, -)$
$(3, 4)$	$(0, -)$	$(-, +)$
$(3, 5)$	$(0, +)$	$(+, -)$

Reviewer 1

This chart is a cool way to check that two triangles just share a vertex. For what it is worth, I did something similar in a recent paper I wrote which solved the minimal vertex problem for origami tori. (This paper appeared recently in the PNAS.)

for $0 < \varepsilon < 1/4$. For example, for the pair $(0, 2)$ one line direction is $(-\varepsilon, \varepsilon/2, -2\varepsilon^2)$, with cone coordinates $(-2\varepsilon, \varepsilon/2)$ and $(\varepsilon(2\varepsilon - 3)/2, \varepsilon)$. The other entries are obtained by the same two-by-two linear calculation; zeros occur only where displayed. Call a sign pair non-negative (respectively non-positive) when both of its entries are so. In no row are the two sign pairs both non-negative or both non-positive. Therefore neither direction of the intersection line lies in both closed tangent cones, and the two closed triangles meet only at their common vertex. This proves the model is embedded. Applying the affine map which sends A_i to E_i gives the stated points $C_i(\varepsilon)$, and they tend to the side midpoints as $\varepsilon \rightarrow 0$. $\square$

Lemma 4 (realizations above $\sqrt{3}$). *Every $\beta > \sqrt{3}$ is realized.*

Proof. Put $\delta = \beta/3$. Choose a with

$$\max\{2/\sqrt{3}, \sqrt{1+\delta^2}\} < a < 2\delta;$$

such an a exists because $\delta > 1/\sqrt{3}$.

Triangulate M_β in the fundamental rectangle $[0, \beta] \times [0, 1]$, with $(0, y) \sim (\beta, 1 - y)$. Put

$$\begin{aligned} A_0 &= (0, 0), & B_0 &= (\delta, 0), & A_1 &= (2\delta, 0), & B_1 &= (\beta, 0), \\ B_1 &= (0, 1), & A_2 &= (\delta, 1), & B_2 &= (2\delta, 1), & A_0 &= (\beta, 1). \end{aligned}$$

Split the three rectangles of width δ by the diagonals A_0A_2 , A_1A_2 , and A_1A_0 , respectively. Equivalently, the six faces are

$$(A_i, B_i, A_{i+2}), \quad (B_i, A_{i+1}, A_{i+2}), \quad i = 0, 1, 2$$

(indices mod 3). This is a clean triangulation of M_β , and its lift to Σ is G_β -invariant.

Use Lemma 3 with side length a , and choose ε so small that the following strict inequalities, which hold in the limit $\varepsilon \rightarrow 0$, still hold:

$$|E_i C_i| < \delta, \quad |C_i E_{i+1}| < \delta, \quad |C_i E_{i+2}| > 1.$$

Also $|E_i E_j| = a > \sqrt{1+\delta^2}$. Map the six vertices A_i, B_i to E_i, C_i , respectively, and extend affinely over each of the six triangles. Lemma 3 gives an embedding $F : M_\beta \rightarrow \mathbb{R}^3$.

Each ridge in the domain has length δ , and its image has length $< \delta$. Each non-boundary edge is of one of two types: either it has domain length 1, and image length $|C_i E_{i+2}| > 1$, or it has domain length $\sqrt{1+\delta^2}$, and image length $a > \sqrt{1+\delta^2}$. Thus F is a squeeze map on every triangle. Lifting F to Σ gives a continuous map invariant under G_β ; since F is an embedding of the quotient, the lifted map satisfies condition (2). Hence β is realized. $\square$

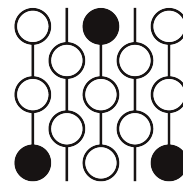
Proposition 1 shows that realized numbers are all $> \sqrt{3}$, while Lemma 4 realizes every $\beta > \sqrt{3}$. Therefore the infimum of the set of realized numbers is $\sqrt{3}$.

Review of Submission 2A

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The solution is entirely correct, and quite well written at that.

The AI quickly establishes that the image is a triangulation of a Möbius band. It then proves the existence of T -patterns and uses this to get the correct lower bound.

For the realization/lower bound, the AI produces a novel triangulation, and argues with great rigor and clarity first that the claimed realization is an embedding, and then that it is a squeeze map with desired aspect ratio.

Novelty. This novelty report shares structure with the novelty report for the other correct solution I refereed. There is no canonical place to put the information common to both, so common information is repeated.

I want to discuss three lemmas where correct solutions varied, and how this solution compares to the others. Then there is the citation issue.

All correct solutions showed that squeeze maps lengthen cevians to the distinguished vertex. All three AI solutions did this by writing down the change in length function and noting that it is a concave-down parabola. The human solution, on the other hand, showed that squeeze maps decrease the distinguished angle, and got cevian length increase from that.

To prove the T -pattern lemma, one can set things up in the right way to reduce to Borsuk-Ulam, or one can directly prove a Borsuk-Ulam-like statement for a slightly different setup. One of the AI solutions, the human solution, and old versions of Schwartz's Möbius bands paper opt for the latter approach. The other two correct solutions, including this one, and the updated version of Schwartz's Möbius bands paper reduce directly to Borsuk-Ulam.

For the realization half, there were two main philosophies. The first was to construct an explicit triangulation, write out the coordinates for all the faces in terms of ϵ , and show everything works out. The other philosophy was to closely approximate a smooth realization. This solution went explicit, employing a 6 triangle triangulation.

Notable: This triangulation was novel to the whole referee team; the human solution and the initial ideas I had in mind when seeing the problem required more triangles or were much nastier. I am impressed at the level of geometric, spacial intuition the model carries – it was able to describe with great accuracy and detail how every piece of its non-obvious construction fit together.

Attribution. The overarching approach by this model is to adapt Schwartz’s Möbius bands paper to a different category. Despite this, the solution never once mentions, alludes to, or cites Schwartz’s work. Proposition 1, which one might call the “optimization” or “geometry” part of the problem, is particularly egregious in its plagiarism. As discussed in the inline comments, it borrows a suspicious amount of notation, the naming for which is motivated by the orientation of a figure featured in Schwartz’s paper. In particular, you would not expect an independent solution, even by someone familiar with the work, to exhibit this common notation.

While the proof is entirely correct, and so will formally receive a “minor revisions” recommendation, a submission of this work by a human to a journal could only be interpreted as willful plagiarism, and would ideally result in a blacklisting.

2 Recommendation

Note that the second half of this solution features original, insightful, creative work, at a level not seen in any of the other AI solutions.

Formally, in accordance with the rubric, this solution receives a “publishable with minor revisions” recommendation, because the only error is a failure of attribution, and this would be easily corrected by citing Schwartz’s prior work.

That said, in the real world this sort of failure of attribution is unacceptable, and would be entirely disqualifying.

3 Submission with inline comments

The infimum is $\sqrt{3}$

Let $M_\beta = \Sigma / \langle G_\beta \rangle$. This is a compact flat Mobius band, with boundary a circle of length 2β . If β is realized, condition (2) says exactly that the given map is constant on the G_β -orbits and separates distinct orbits. Hence it descends to a continuous injection $F : M_\beta \rightarrow \mathbb{R}^3$, and this injection is an embedding because M_β is compact and $\mathbb{R}^3$ is Hausdorff. The locally finite invariant triangulation descends to a locally finite triangulation of the compact surface M_β , hence to a finite one.

In a clean triangle, call the boundary edge the ridge and the opposite vertex the apex. A bend is a segment in the triangle joining the apex to a point of the ridge.

Reviewer 2

This is a very strong/well-written start. Concise and precise, and sets up some useful definitions with intuitive names.

Lemma 1 (bends lengthen). *Let L be the affine map on one clean triangle. If P is any point of the ridge and C is the apex, then*

$$|L(P) - L(C)| > |P - C|.$$

Moreover every non-degenerate boundary subarc is strictly shortened by F .

Proof. Let the ridge endpoints be A, B , and put $P_s = (1 - s)A + sB$. The function

$$q(s) = |L(P_s) - L(C)|^2 - |P_s - C|^2$$

satisfies

$$q''(s) = 2(|L(A) - L(B)|^2 - |A - B|^2) < 0,$$

because the ridge is shortened. Since the two side edges are lengthened, $q(0) > 0$ and $q(1) > 0$. Concavity gives $q(s) > 0$ for $0 \leq s \leq 1$. On each boundary edge the affine map has constant Lipschitz factor < 1 ; an arbitrary boundary subarc is split at the finitely many boundary vertices it meets, and the strict inequalities add. $\square$

Lemma 2 (a polygonal T -pair). *For $F(M_\beta)$ there are two bends whose image segments lie on perpendicular intersecting lines.*

Proof. The bends form a circle. Indeed, every bend meets the center circle $\{y = 1/2\}/\langle G_\beta \rangle$ once, and every point of the center circle lies in a unique bend; on a common non-boundary edge the two adjacent triangles give the same limiting bend.

Parametrize this bend circle by $u \in \mathbb{R}/\mathbb{Z}$, lift to $u \in \mathbb{R}$, and choose the image direction $e(u)$ from the lower side of the lifted strip to the upper side. Let $m(u)$ be the image midpoint. The previous lemma makes every image bend non-degenerate, and the finite polygonal structure gives continuous functions e, m . Going once around the Mobius band reverses the chosen orientation, so

$$e(u + 1) = -e(u), \quad m(u + 1) = m(u).$$

For $0 < r < 1$ define

$$\Phi(u, r) = \left(e(u) \cdot e(u + r), (m(u) - m(u + r)) \cdot (e(u) \times e(u + r)) \right).$$

It is periodic in u . As $r \rightarrow 0$ it tends to $(1, 0)$, and as $r \rightarrow 1$ it tends to $(-1, 0)$. Thus it extends continuously to the two-point compactification of $S^1 \times (0, 1)$, obtained by collapsing the two boundary circles of $S^1 \times [0, 1]$; this compactification is S^2 .

The involution $\iota(u, r) = (u + r, 1 - r)$ interchanges the two bends. Under the coordinates

$$(\sin \pi r \cos 2\pi(u + r/2), \sin \pi r \sin 2\pi(u + r/2), \cos \pi r)$$

on the compactified cylinder, ι is the antipodal map. From $e(u + 1) = -e(u)$ and $m(u + 1) = m(u)$ one checks that $\Phi \circ \iota = -\Phi$. Borsuk–Ulam gives a zero of Φ .

At such a zero the two direction vectors are perpendicular. The second coordinate says that the vector joining the two midpoints lies in the plane spanned by those directions; hence the two supporting lines are coplanar. Since they are not parallel, they intersect. $\square$

Proposition 1. *Every realized β satisfies $\beta > \sqrt{3}$.*

Proof. Choose the two bends from Lemma 2. Their image segments have disjoint relative interiors, because F is an embedding. Let O be the intersection of their supporting lines. At least one image segment does not contain O in its relative interior; call that bend B , and call the other one T .

Lift T to a segment from $(0, 0)$ to $(t, 1)$. Replacing x by $-x$, if necessary, we may assume $t \geq 0$. In fact $t < \beta$: if $t > \beta$, this lifted segment meets its G_β -translate in its interior, and if $t = \beta$, its endpoints are G_β -equivalent. Either case contradicts the fact that the image of a bend is a non-degenerate affine segment and that F is embedded on the quotient.

Cutting M_β along T gives two boundary arcs H, D of lengths

$$\text{length}(H) = \beta - t, \quad \text{length}(D) = \beta + t. \quad (1)$$

Indeed the upper endpoint $(t, 1)$ has boundary coordinate $t + \beta$ modulo 2β . Let $T^* = F(T)$. Since $H^* = F(H)$ joins the endpoints of T^* , while boundary arcs are shortened and bends are lengthened,

$$\beta - t = \text{length}(H) > \text{length}(H^*) \geq |T^*| > \sqrt{1 + t^2}. \quad (2)$$

The line of B^* is perpendicular to the line of T^* . Since O is not in the relative interior of B^* , one endpoint v^* of B^* has distance from the line of T^* at least $|B^*|$. The domain length of B is at least 1, so Lemma 1 gives $|B^*| > 1$. Let A be the boundary arc, either H or D , which contains the corresponding boundary endpoint v .

The image arc A^* goes from one endpoint of T^* to the other through v^* . If a point has perpendicular distance h from a line segment of length L , the sum of its distances to the two endpoints is at least $\sqrt{L^2 + 4h^2}$, with equality only over the midpoint. Here $L = |T^*| > \sqrt{1 + t^2}$ and $h > 1$, so

$$\text{length}(A) > \text{length}(A^*) \geq (\text{that sum}) > \sqrt{5 + t^2}. \quad (3)$$

I found (3) entertaining.

Reviewer 2

If $A = H$, then $D \geq H$ and (3) gives

$$2\beta = H + D > 2\sqrt{5 + t^2} > 2\sqrt{3}.$$

Thus assume $A = D$. From (1)–(3),

$$2\beta = H + D > \sqrt{1 + t^2} + \sqrt{5 + t^2}, \quad (4)$$

and also

$$2\beta = 2D - 2t > 2\sqrt{5 + t^2} - 2t. \quad (5)$$

Let

$$a(t) = \sqrt{1 + t^2} + \sqrt{5 + t^2}, \quad b(t) = 2\sqrt{5 + t^2} - 2t.$$

On $[0, \infty)$, a is increasing and b is decreasing, and

$$a(1/\sqrt{3}) = b(1/\sqrt{3}) = 2\sqrt{3}.$$

Hence $\max(a(t), b(t)) \geq 2\sqrt{3}$ for all $t \geq 0$. Equations (4)–(5) imply $2\beta > 2\sqrt{3}$. $\square$

It remains to construct realizations for all larger β .

Reuses the notation t, T, B, H, D of Schwartz's paper. Does not cite Schwartz's paper in any way, despite the plethora of ideas taken from it.

Lemma 3 (an embedded six-triangle model). *Let $E_0E_1E_2$ be an equilateral triangle of side a . For every sufficiently small $\varepsilon > 0$ there are points $C_i(\varepsilon)$ tending to the midpoint of E_iE_{i+1} (indices mod 3) such that the six affine triangles*

$$(E_i, C_i, E_{i+2}), \quad (C_i, E_{i+1}, E_{i+2}), \quad i = 0, 1, 2,$$

form an embedded polyhedral Mobius band with boundary $E_0C_0E_1C_1E_2C_2E_0$.

Proof. Embeddedness is preserved by nonsingular affine changes of coordinates, so it is enough to verify one model. Take

$$\begin{aligned} A_0 &= (0, 0, 0), & A_1 &= (1, 0, 0), & A_2 &= (0, 1, 0), \\ B_0 &= (1/2, 0, \varepsilon), & B_1 &= (1/2 - \varepsilon, 1/2, -2\varepsilon), & B_2 &= (\varepsilon, 1/2, 0), \end{aligned}$$

where $0 < \varepsilon < 1/4$, and take the six faces

$$\begin{aligned} \Delta_0 &= A_0B_0A_2, & \Delta_1 &= B_0A_1A_2, & \Delta_2 &= A_1B_1A_0, & \Delta_3 &= B_1A_2A_0, \\ \Delta_4 &= A_2B_2A_1, & \Delta_5 &= B_2A_0A_1. \end{aligned}$$

The links of the vertices are intervals. For instance

$$\text{lk}(A_i) = B_i - A_{i+2} - B_{i+1} - A_{i+1} - B_{i+2}, \quad \text{lk}(B_i) = A_i - A_{i+2} - A_{i+1}$$

By the link of a vertex v , it means the graph whose vertices are those adjacent to v , and whose edges are the opposite sides of triangles containing v . The $X - Y - Z$ notation does not indicate a formal sum of points; rather the “ $-$ ” signs are meant to indicate *edges in a path graph*. The notation is cute in hindsight but very cryptic on first read.

(indices mod 3, and reversal of a path is immaterial). Thus the abstract complex is a connected compact surface. Its single boundary cycle is $A_0B_0A_1B_1A_2B_2A_0$, and $\chi = 6 - 12 + 6 = 0$; hence, by the classification of compact surfaces, it is a Mobius band once its geometric realization is embedded.

It remains to check embeddedness. The face normals are

$$\begin{aligned} n_0 &= (-\varepsilon, 0, 1/2), & n_1 &= (\varepsilon, \varepsilon, 1/2), & n_2 &= (0, 2\varepsilon, 1/2), \\ n_3 &= (2\varepsilon, 0, 1/2 - \varepsilon), & n_4 &= (0, 0, 1/2 - \varepsilon), & n_5 &= (0, 0, 1/2). \end{aligned}$$

For the edge-sharing pairs $(0, 1), (0, 3), (1, 4), (2, 3), (2, 5)$ these normals are not proportional, so the two planes meet in the line of the common edge and the two triangles meet exactly in that edge. The remaining edge-sharing pair is $(4, 5)$. These two triangles are coplanar in $z = 0$, but their third vertices lie on opposite sides of their common edge B_2A_1 , since

$$\det(A_1 - B_2, A_2 - B_2) = \frac{1 - 2\varepsilon}{2} > 0, \quad \det(A_1 - B_2, A_0 - B_2) = -\frac{1}{2} < 0.$$

Therefore $\Delta_4 \cap \Delta_5 = B_2 A_1$.

The other nine pairs of faces share only one vertex. For each of them, the normals listed above are not proportional, so the two planes meet in a line through that vertex. Expressing one direction of this line in the two edge bases at the shared vertex gives the following sign table:

(i, j)	Δ_i	Δ_j
(0, 2)	(-, +)	(-, +)
(0, 4)	(-, 0)	(-, +)
(0, 5)	(0, -)	(-, +)
(1, 2)	(-, +)	(+, -)
(1, 3)	(+, -)	(-, +)
(1, 5)	(0, +)	(+, -)
(2, 4)	(0, -)	(+, -)
(3, 4)	(0, -)	(-, +)
(3, 5)	(0, +)	(+, -)

for $0 < \varepsilon < 1/4$. For example, for the pair (0, 2) one line direction is $(-\varepsilon, \varepsilon/2, -2\varepsilon^2)$, with cone coordinates $(-2\varepsilon, \varepsilon/2)$ and $(\varepsilon(2\varepsilon - 3)/2, \varepsilon)$. The other entries are obtained by the same two-by-two linear calculation; zeros occur only where displayed. Call a sign pair non-negative (respectively non-positive) when both of its entries are so. In no row are the two sign pairs both non-negative or both non-positive. Therefore neither direction of the intersection line lies in both closed tangent cones, and the two closed triangles meet only at their common vertex. This proves the model is embedded. Applying the affine map which sends A_i to E_i gives the stated points $C_i(\varepsilon)$, and they tend to the side midpoints as $\varepsilon \rightarrow 0$. $\square$

Reviewer 2

Lemma 3 and its proof are very impressive. Drawing on a blackboard confirms that all the coordinates, triangles, etc. described behave as claimed, and the proof does a great job proving everything rigorously without reference to visualization or figures.

Lemma 4 (realizations above $\sqrt{3}$). *Every $\beta > \sqrt{3}$ is realized.*

Proof. Put $\delta = \beta/3$. Choose a with

$$\max\{2/\sqrt{3}, \sqrt{1 + \delta^2}\} < a < 2\delta;$$

such an a exists because $\delta > 1/\sqrt{3}$.

Triangulate M_β in the fundamental rectangle $[0, \beta] \times [0, 1]$, with $(0, y) \sim (\beta, 1 - y)$. Put

$$\begin{aligned} A_0 &= (0, 0), & B_0 &= (\delta, 0), & A_1 &= (2\delta, 0), & B_1 &= (\beta, 0), \\ B_1 &= (0, 1), & A_2 &= (\delta, 1), & B_2 &= (2\delta, 1), & A_0 &= (\beta, 1). \end{aligned}$$

Split the three rectangles of width δ by the diagonals $A_0 A_2$, $A_1 A_2$, and $A_1 A_0$, respectively. Equivalently, the six faces are

$$(A_i, B_i, A_{i+2}), \quad (B_i, A_{i+1}, A_{i+2}), \quad i = 0, 1, 2$$

(indices mod 3). This is a clean triangulation of M_β , and its lift to Σ is G_β -invariant.

Use Lemma 3 with side length a , and choose ε so small that the following strict inequalities, which hold in the limit $\varepsilon \rightarrow 0$, still hold:

$$|E_i C_i| < \delta, \quad |C_i E_{i+1}| < \delta, \quad |C_i E_{i+2}| > 1.$$

Also $|E_i E_j| = a > \sqrt{1 + \delta^2}$. Map the six vertices A_i, B_i to E_i, C_i , respectively, and extend affinely over each of the six triangles. Lemma 3 gives an embedding $F : M_\beta \rightarrow \mathbb{R}^3$.

Each ridge in the domain has length δ , and its image has length $< \delta$. Each non-boundary edge is of one of two types: either it has domain length 1, and image length $|C_i E_{i+2}| > 1$, or it has domain length $\sqrt{1 + \delta^2}$, and image length $a > \sqrt{1 + \delta^2}$. Thus F is a squeeze map on every triangle. Lifting F to Σ gives a continuous map invariant under G_β ; since F is an embedding of the quotient, the lifted map satisfies condition (2). Hence β is realized. $\square$

Proposition 1 shows that realized numbers are all $> \sqrt{3}$, while Lemma 4 realizes every $\beta > \sqrt{3}$. Therefore the infimum of the set of realized numbers is $\sqrt{3}$.